Roll No. ....

TPE–007

B. TECH. (ECE) (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

SEMICONDUCTOR DEVICES AND MODELING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Explain the variation of electron concentration with temperature for *n*-type semiconductor. (CO1)

OR

(b) Explain the effect of gate body voltage of two terminal MOS structure on : (CO1)

(i) Flatband condition

(ii) Accumulation

(iii) Depletion and inversion

2. (a) Calculate the flatband voltage for a p-type body with $N_A = 10^{18}$ cm^{-3}, a SiO_2 insulator with a thickness t_{ox} = 2 nm, and n-type polysilicon gate with $N_D = 10^{20}$ cm^{-3}. The interface charge $Q_0 = 10^{-8}$ C/cm^2 and $\phi_{ms} = -1.036$ V. (CO1)

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OR

(b) Explain three terminal MOS structure with proper band diagram.

(CO1)

3. (a) Write short notes on the following : (CO1, CO2)

(i) Body effect of MOS Transistor

(ii) Drain Induced Barrier Lowering (DIBL)

OR

(b) A step graded Ge diode, having $N_D = 500\ N_A$, acceptor impurities are 2 impurity atoms for every 10^8 atoms at room temperature. Find the barrier potential. Assume $n_i = 2.5 \times 10^{13}/cm^3$. Total number of Ge atoms $= 4.4 \times 10^{22}/cm^3$. (CO1, CO2)

4. (a) Write short notes on the following : (CO2)

(i) Channel Length Modulation

(ii) Narrow Channel Effects

OR

(b) Explain flatband voltage in detail along with expressions. (CO2)

5. (a) Explain in detail about formation of energy band. (CO2)

OR

(b) Determine the voltage gain, current gain, input impedance and output impedance for common-drain amplifier. (CO2)

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